

MACHINE LEARNING

CORE CONCEPTS HANDBOOK

Training Accuracy & Testing Accuracy

Accuracy

Accuracy measures how many predictions are correct out of total predictions.

Formula:

$$\text{Accuracy} = (\text{Correct Predictions}) / (\text{Total Predictions})$$

In classification:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Range:
0 to 1 (or 0% to 100%)

Training Accuracy

Training Accuracy is the accuracy of the model on the training dataset — the same data that was used to train the model.

It tells us: How well the model has learned patterns from training data.

Example

Suppose:

Training dataset has 100 samples.

Model correctly predicts 95 of them.

$$\text{Training Accuracy} = 95 / 100 = 0.95 \text{ (95\%)}$$

High training accuracy does NOT mean model is good.

Because:
Model might simply memorize training data.

Testing Accuracy

Testing Accuracy is the accuracy of the model on unseen data (test dataset).

It tells us: How well the model generalizes to new data.

Example

Suppose:

Testing dataset has 40 samples.

Model correctly predicts 28.

$$\text{Testing Accuracy} = 28 / 40 = 0.70 \text{ (70\%)}$$

Importance of Testing Accuracy

Because in real life:
Model will see new data, not training data.

So testing accuracy shows real performance.

Relationship Between Training & Testing Accuracy

Case 1: Overfitting

Example:

Training Accuracy = 98%

Testing Accuracy = 65%

Model performs extremely well on training data

But poorly on new data

- Model memorized training data
- High Variance
- Overfitting

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Case 2: Underfitting

Example:

Training Accuracy = 60%

Testing Accuracy = 58%

Model performs poorly on both training and testing data.

- Model is too simple
- High Bias
- Underfitting

Case 3: Good Fit

Example:

Training Accuracy = 88%

Testing Accuracy = 85%

Model performs well on training

Model performs similarly well on test

- Good generalization
- Balanced bias & variance

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Summary Table

Scenario	Training Accuracy	Testing Accuracy	Problem Type
Low – Low	Low	Low	Underfitting
High – Low	High	Low	Overfitting
High – High	High	High	Good Model

Training Accuracy is Usually Higher :

Because:

Model already saw training data during learning.

So it naturally performs better on it.

Testing accuracy is usually slightly lower.

That is normal.

Acceptable Difference Between Training & Testing Accuracy

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Small difference (2%–5%) → Normal

Moderate difference (5%–10%) → Check model

Large difference (>15%) → Likely overfitting

Training vs Validation vs Testing Accuracy

In real ML projects we use 3 datasets:

1. Training Set → Used to train model
2. Validation Set → Used to tune hyperparameters
3. Testing Set → Used for final evaluation

Validation accuracy helps detect overfitting before final testing.

Example

```
model.fit(X_train, y_train)

train_acc = model.score(X_train, y_train)
test_acc = model.score(X_test, y_test)

print("Training Accuracy:", train_acc)
print("Testing Accuracy:", test_acc)
```

Bias

Bias is the error caused by overly simple assumptions in the model.

High bias means:

Model cannot capture true pattern.

High Bias → Underfitting

Example

Fitting straight line to curved data.

Model assumption is too strong.

Variance

More

Labels

Inbox

Starred

Snoozed

Sent

Drafts

Purchases

More

Labels

1,168

7

22

More

Labels

Inbox

Starred

Snoozed

Sent

Drafts

Purchases

More

Labels

1,168

7

22

Variance is the error caused by model being too sensitive to small fluctuations in training data.

High variance means:
Model changes drastically with small changes in data.

High Variance → Overfitting

Example

Deep decision tree that perfectly fits training set.

Remove one training sample → model changes completely.

Bias-Variance Tradeoff

There is always a tradeoff:

- Increasing complexity ↓ Bias but ↑ Variance
- Decreasing complexity ↑ Bias but ↓ Variance

Goal:
Find balance where total error is minimum.

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INNOVATION & CREATIVITY

Enriching Your Better Tomorrow

Practical Detection in Code

If:

Training Accuracy = 98%
Testing Accuracy = 65%

→ Overfitting

If:

Training Accuracy = 60%
Testing Accuracy = 58%

→ Underfitting

Important Ranges

Accuracy: 0-1
Bias: Low is better
Variance: Low is better
Good model: Moderate bias + Moderate variance

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